

Assignment - 3

DBMS

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SECTION - A

Ques- What is SQL? Why is it important in DBMS?

Ans- SQL (Structured Query Language) is the standard language used to communicate with relational databases. It is used to create tables, insert data, update records, delete data, and retrieve information as per user requirement. SQL is important because :-

(i) It provides a simple and efficient way to manage large volume of structure data.

(ii) It ensures data accuracy, supports security through user permission, and allows multiple user to access the database simultaneously.

(iii) Almost, all modern application like banking system, hospital management, e-commerce websites, and university portals rely on SQL to handle their data operations.

Ques- What is the difference between DDL & DML?

Ans- DDL (Data Definition Language)

→ It deals with defining & modifying the structure of database objects such as tables and schemas. Commands include DROP, TRUNCATE, ALTER, CREATE. These commands change how the database is organised.

→ DML (Data Manipulation Language)

It deals with operations on the data stores inside the database. Commands include INSERT, UPDATE, DELETE and SELECT. These commands modify or retrieve the actual data.

Thus, DDL manages structure, while DML manages content.

Q3- What is a JOIN? Explain its types.

Ans- A JOIN is an SQL operation used to combine data from two or more tables based on a common related column, usually a primary key and foreign key. Since data in relational databases is stored in normalized form across multiple tables, JOIN is necessary to retrieve complete and meaningful information.

→ Types of JOIN:-

(i) INNER JOIN:-

Returns only the matching records from both tables.

(ii) LEFT JOIN:-

Returns all records from the left table and matched records from the right table.

(iii) RIGHT JOIN:-

Returns all records from the right table and matched records from the left table.

(iv) FULL JOIN:-

Returns all records from both tables whether matched or not.

Ques 4- What is GROUP BY and HAVING clause?

Ans 4- GROUP BY is used to group rows that have the same values in specific columns into summary rows. It is generally used to apply conditions on group data after GROUP BY aggregate functions like COUNT, SUM, AVG, MAX and MIN to produce summarized results.

HAVING is used to apply conditions on grouped data after GROUP BY is executed. It filters groups rather than individual rows.

GROUP BY helps in summarizing data, while HAVING helps in filtering that summarized data.

Ques 5- What is a Subquery?

Ans 5- A subquery is a query inside another query, enclosed in parentheses. The inner query is executed first and its result is used by the outer query. Subqueries are useful when a query depends on the result of another query, such as finding students who scored more than the average marks.

Ques 6- What is relational algebra?

Ans 6- Relational algebra is a procedural query language that forms the theoretical foundation of SQL. It consists of operations like select, project, join, union, Intersection, Difference and Rename. These operations describe step-by-step how to retrieve data from relations (tables). It helps in understanding how queries work internally in DBMS.

Ques- Difference between WHERE and HAVING.
 Ans- WHERE filters individual rows before grouping and can be used with aggregate functions.
 HAVING filters grouped results after GROUP BY and is useful with aggregate function.

SECTION-B

Ques- Write an SQL query to retrieve all students with marks > 75.

Student

St-Name	St-marks	St-ID
A	78	001
B	90	002
C	60	003
D	72	004

SELECT *
 FROM STUDENT
 WHERE St-marks > 75;

Output:- 78
 90

Explanation:-
 The WHERE clause checks each rows and return only those students whose marks are greater than 75. So it filters data before any grouping and aggregation.

Ques 2 - Write a query ^{using} to join to combine student and course table.

Ans 2 -

Course

Course-Name	Course-ID
DBMS	001
C++	002
Java	003
Python	004

Student

St-course-id	St-name
DBMS	A
C++	B
Java	C
Python	D

```
SELECT St_name, Course.Course-name
FROM STUDENT
```

```
INNER JOIN Course
```

```
ON St-course-id = Course.Course-id.
```

student & course

Output -

St-course-id	St-course	St-name
001	DBMS	A
002	C++	B
003	Java	C
004	python	D

Explanation - This query uses inner join to join student and course table through the common field course name. It returns table after join, and shows which student enrolls in which course with its id.

Ques 3 - Write a query using GROUP BY to count student per department.

Ans 3-

Student	St Id	St-dept.
St_name		
A	01	SOFT
B	02	SOM&C
C	03	SOASSOFT
D	04	BHMA-SOFT

SELECT department, COUNT(*) AS total_students
FROM Student
GROUP BY department;

Explanations-

GROUP BY arranges students based on department and COUNT() calculates how many students belong to each department, producing summarized results.

Section-C

Ques 1- A company database stores employees and department data. Write a query to find employees in a specific department.

Ans 1-

Department	Employee
Dept_name	Employee_id
HR	001
Manager	002
Assistant	003
Staff	004

Pg No. _____

```

SELECT Employee-name
FROM Employee
INNER JOIN Department
ON Employee-dept_id = Dept-dept_id
WHERE Department-dept-name = 'HR';
  
```

→ Output:- 01 (dept-id).

→ Explanation:-

The join connected both tables using dept_id. The where clause filters employee whose id is on that department name.

Ques 2 - An e-commerce system stores orders. Retrieve total sales per consumer.

Ans 2 -

Orders

Cust-name	Cust_id	sale	Amount
A	01	2	50,000
B	02	3	60,000
C	03	4	70,000
D	04	1	40,000

```

Select customer_id, SUM(amount) AS total-Sales
FROM ORDERS
GROUP BY customer_id;
  
```

→ Explanation:-

Orders are groups by customer, and SUM() calculates the purchase amount for each customer, which is useful for sale analysis.

Ques 3- A university database needs top 3 students, write a query

Ans 3-

Student		
St-name	St marks	St id
A	50%	001
B	90%	002
C	95%	003
D	99%	004
E	80%	005
F	75%	006

```
SELECT name, marks
FROM Student
ORDER BY marks DESC
LIMIT 3;
```

→ Output - B, C, D.

→ Explanation -

ORDER BY sorts student by highest marks first, and LIMIT restricts the output to the top 3 students.

Section - D

Ques 1- Write an SQL query for any real-life system and explain logic.

Ans 1-

Patient

Patient Id	name	age	disease
P101	A	45	fever
P102	B	67	Diabetes
P103	C	72	BP
P104	D	56	flu flu

→ Output: B C D
67, 72, 65.

~~SELECT name, age~~
SELECT age, name
FROM Patient
WHERE age > 60;

→ insight - This query helps hospitals quickly identify senior citizens who may require priority treatment or special care.

Ques 2 - Write a Join query & mention one use case.

Ans 2 -

Customer

Cust_id	name	city
C1	A	Delhi
C2	B	Mumbai
C3	C	Pune

Orders

Order_id	Cust_id	amount
101	C1	2800
102	C2	1800
103	C3	3200

SELECT Orders.order_id, Customer.name
FROM Orders

JOIN Customer

ON Orders.customer_id = Customer.customer_id;

→ Outputs -

order-id	name
101	A
102	B
103	C

→ Use Case - This shows which customer placed which order, helping in order tracking and Customer service.

Que 3- Write a GROUP BY query and ^{explain} its output.
Ans 3-

Employee

emp-id	name	department
E1	A	HR
E2	B	IT
E3	C	HR
E4	D	Sales
E5	E	IT

SELECT department, Count(*)
FROM Employee
GROUP BY department;

→ Outputs - HR = 2, IT = 2, Sales = 1.

Explanation -

This query provides the count of employees in each department, which is helpful for management and resource planning.